

Statistical Learning

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Outline

- Statistical learning
- Machine learning

Introduction

- Although structuring and visualizing data are important aspects of data science, the main challenge lies in the **mathematical analysis of the data**.
- When the goal is to interpret the model and quantify the uncertainty in the data, this analysis is usually referred to as **statistical learning**.
- **In contrast**, when the emphasis is on making predictions using large-scale statistical data, then it is common to speak about **machine learning or data mining**

Goals for modeling data

- There are two major goals for modeling data:
 - to accurately predict some future quantity of interest, given some observed data, and
 - to discover unusual or interesting patterns in the data.
- To achieve these goals, one must rely on knowledge from three important pillars of the mathematical sciences.
 - Function approximation.
 - Optimization.
 - Probability and Statistics.

Function approximation

- Building a **mathematical model** for data usually means understanding how one data variable depends on another data variable.
 - The most natural way to represent the relationship between variables is via a **mathematical function or map**.
- We usually assume that this mathematical **function is not** completely known but can be approximated well given enough computing power and data.
- Thus, **data scientists** have to understand how best to approximate and represent functions using the least amount of computer processing and memory

Optimization

- Given a class of mathematical models, we wish to find the **best possible model** in that class.
- This requires some kind of **efficient search** or **optimization** procedure.
- The optimization step can be viewed as a process of fitting or calibrating a function to observed data.
- This step usually requires **knowledge of optimization algorithms** and efficient computer coding or programming.

Probability and Statistics.

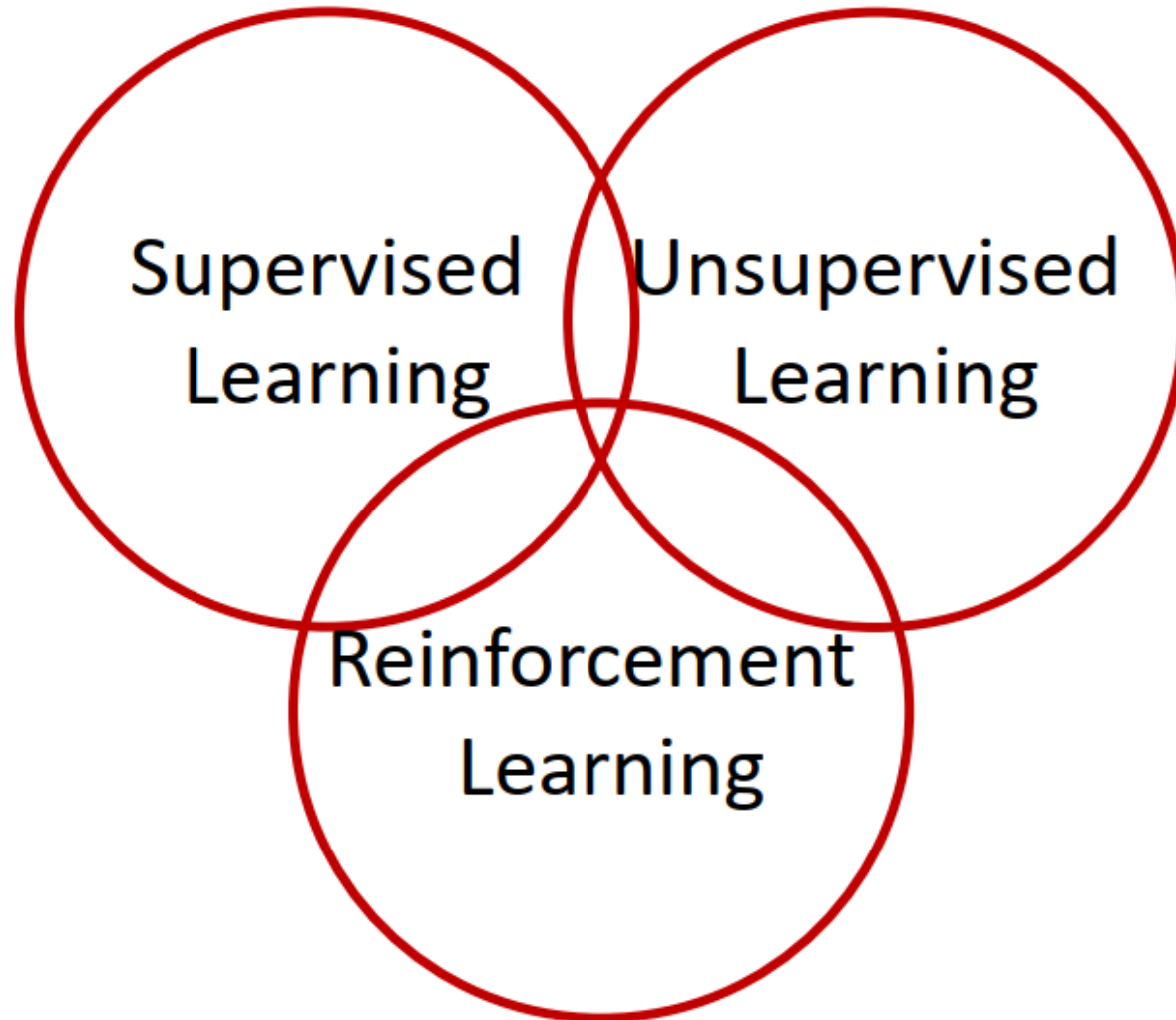
- In general, the **data** used to **fit** the model is viewed as a realization of a random process or numerical vector, whose probability law determines the accuracy with which we can predict future observations.
- Thus, in order to quantify the uncertainty inherent in making predictions about the future, and the sources of error in the model, **data scientists** need a firm grasp of probability theory and statistical inference.

What is machine learning?

- In 1959, the term “machine learning” was first introduced by Arthur Samuel.
- field of study that gives computers the ability to learn without being explicitly programmed
- It uses statistical methods to give computer the ability to "learn" from data, without being explicitly programmed.



Taxonomy of Machine Learning



Types of Machine Learning

- Supervised (inductive) learning
 - Given: training data + desired outputs (labels)
- Unsupervised learning
 - Given: training data (without desired outputs)
- Semi-supervised learning
 - Given: training data + a few desired outputs
- Reinforcement learning
 - Rewards from sequence of actions

Supervised and Unsupervised Learning

➤ Feature and Response

- Given an input or feature vector x , one of the **main goals of machine** learning is to predict response an output or response variable y .
- For example, x could be a digitized signature and y a binary variable that indicates whether the signature is genuine or false

➤ Prediction Function

- The data science attempt at this prediction is encoded in a mathematical prediction function g , called the **prediction function**, which takes as an input x and outputs a guess $g(x)$ for y (denoted by y , for example).
- In a sense, g encompasses all the information about the relationship between the variables x and y , excluding the effects of chance and randomness in nature.

Regression & Classification

- In **regression** problems, the response variable y can take any real value (Continuous Value).
- In contrast, when y can only lie in a finite set, say $y \in \{0, \dots, c - 1\}$, then predicting y is conceptually the same as classifying the input x into one of c categories, and so prediction becomes a **classification problem**.

Loss Function

- We can measure the accuracy of a prediction by with respect to a given response y by loss function using some $\text{Loss}(y, \hat{y})$.
- In a **regression** setting the usual choice is the squared error loss $\text{loss}(y, \hat{y})^2$.
- In the case of **classification**, the zero–one (also written 0–1) loss function $\text{Loss}(y, \hat{y}) = 1 \text{ if } \{y \neq \hat{y}\}$ is often used, which incurs a loss of 1 whenever the predicted class by is not equal to the class y .

Error

The word *error* is often used as a measure of distance between a “true” object y and some approximation $\widehat{y}$ thereof. If y is real-valued, the absolute error $|y - \widehat{y}|$ and the squared error $(y - \widehat{y})^2$ are both well-established error concepts, as are the norm $\|y - \widehat{y}\|$ and squared norm $\|y - \widehat{y}\|^2$ for vectors. The squared error $(y - \widehat{y})^2$ is just one example of a loss function.

Risk

- It is unlikely that any mathematical function g will be able to make accurate predictions for all possible pairs $(x; y)$
- **One reason** for this is that, even with the same input x , the output y may be different, depending on chance circumstances or randomness
- For this reason, we adopt a probabilistic approach and assume that each pair $(x; y)$ is the outcome of a random pair $(X; Y)$ that has some joint probability density $f(x; y)$.
- We then assess the **predictive performance** via the expected loss, usually called the **risk**, for g

$$\ell(g) = \mathbb{E} \text{Loss}(Y, g(X)).$$

Risk

- **For example**, in the classification case with zero–one loss function the **risk** is equal to the probability of incorrect classification:

$$\ell(g) = \mathbb{P}[Y \neq g(X)]$$

- In this context, the prediction function g is called a **classifier**
- Given the distribution of $(X; Y)$ and any loss function, we classifier can in principle find the best possible

$$g^* := \operatorname{argmin}_g \mathbb{E} \operatorname{Loss}(Y, g(X))$$

Training Loss

- Given an arbitrary prediction function g , it is typically not possible to compute its risk $\ell(g)$ in

$$\ell(g) = \mathbb{E} \text{Loss}(Y, g(X)).$$

- However, using the training sample T , we can approximate $\ell(g)$ via the empirical (sample average) risk

$$\ell_T(g) = \frac{1}{n} \sum_{i=1}^n \text{Loss}(Y_i, g(X_i)),$$

- which we call the **training loss**.
- The **training loss** is thus an unbiased estimator of the risk (the expected loss) for a prediction function g , based on the training data

Optimal Prediction Function

- To approximate the **optimal prediction function** g^* (the minimizer of the risk $\ell(g)$)
- We first select a suitable **collection of approximating functions** G and then take our learner to be the function in G that minimizes the training loss; that is,

$$g_{\mathcal{T}}^G = \operatorname{argmin}_{g \in G} \ell_{\mathcal{T}}(g).$$

Overfitting

- Poor generalization performance is called **overfitting**

By choosing g a function that predicts the training data exactly (and is, for example, 0 otherwise), the squared-error training loss is zero. Minimizing the training loss is not the ultimate goal!

Generalization Risk

- The prediction accuracy of new pairs of data is measured by the generalization risk of the learner.
- For a fixed training set τ it is defined as $\ell(g_\tau^\mathcal{G}) = \mathbb{E} \text{Loss}(Y, g_\tau^\mathcal{G}(X))$,
 - where $(X; Y)$ is distributed according to $f(x; y)$
- **The generalization risk** is the average loss over all possible pairs $(x; y)$, weighted by the corresponding $f(x; y)$

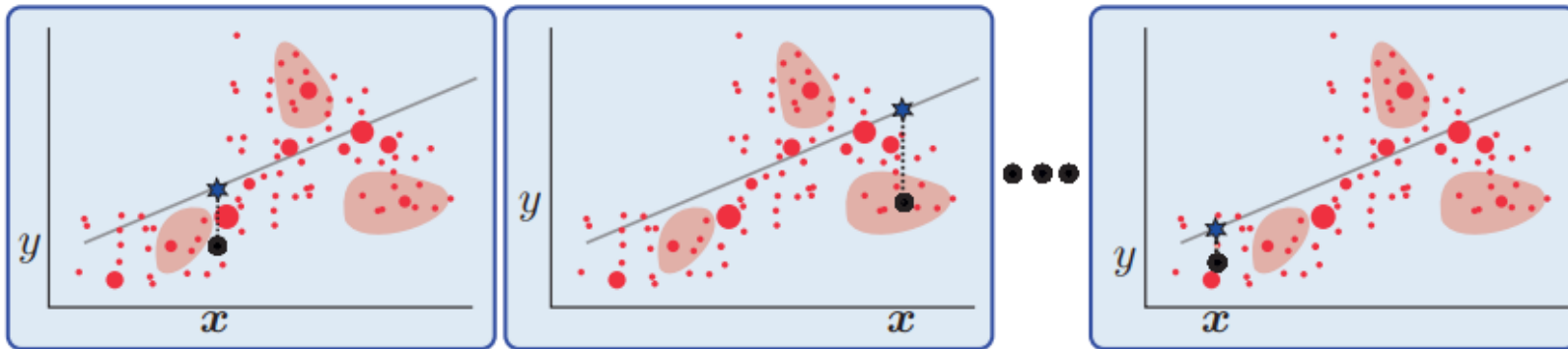


Figure 2.1: The generalization risk for a fixed training set is the weighted-average loss over all possible pairs (x, y) .

Tradeoffs in Statistical Learning

- The art of machine learning in the supervised case is to make the generalization risk or expected generalization risk as small as possible, while using as few computational resources as possible.
- In pursuing this goal, **a suitable class G** of prediction functions has to be chosen. This choice is driven by **various factors**, such as
 - the complexity of the class (e.g., is it rich enough to adequately approximate, or even contain, the optimal prediction function g^* ?),
 - the ease of training the learner via the optimization program,
 - how accurately the training loss estimates the risk within class G ,
 - the feature types (categorical, continuous, etc.)

Thank You!